# Beyond Pairwise GUI Quality Judgments: Multi-Dimensional Evaluation with Modern Multimodal LLMs

Xinyang Xie

Department of Informatics, Technische Universitaet Clausthal, Clausthal-Zellerfeld, Germany.

**Abstract**

Assessing the quality of graphical user interfaces (GUIs) is becoming increasingly important as large language models are used to generate interface designs and frontend code. Existing screenshot-based approaches can support GUI quality judgments, but they remain limited by binary decisions, older models, and simple prompting strategies. This proposal extends that line of work with current multimodal large language models. The evaluation is organized around two complementary dataset tracks: a screenshot-based baseline drawn from existing UI resources, and a requirement-driven track in which interfaces are generated from natural language descriptions. The study combines pairwise comparison with rubric-based assessment across multiple dimensions, including layout quality, visual hierarchy, information organization, perceived usability, and – for generated interfaces – requirement fidelity and interaction quality. Methodologically, the work begins with a literature review of LLM-as-a-judge techniques to identify applicable strategies, followed by a systematic comparison of approaches such as zero-shot prompting, rubric-guided scoring, repeated judging, and order-swapped pairwise comparisons, where the order of candidates is reversed to detect positional bias. A human annotation protocol with agreement analysis provides reference labels. The resulting protocol will be packaged into a benchmark and lightweight leaderboard that can be extended to newly added models. In addition to the static evaluation tracks, the proposal includes a small dynamic evaluation module in which computer-use agents attempt short tasks on selected interfaces. The goal is to determine which evaluation setup aligns best with human judgments and how static quality judgments relate to dynamic interaction performance.

# 1 Introduction and Motivation

The quality of a graphical user interface (GUI) shapes how efficiently users complete tasks and how they perceive a software system. As generative models are increasingly used to produce interface mockups and frontend code, the question of how UI quality can be evaluated becomes more pressing. Importantly, GUI quality is not limited to visual appearance alone: it also concerns whether an interface supports successful interaction and task completion once users begin to navigate it.

UIClip [**?** ] provided an important first step by showing that vision-language models can perform pairwise comparisons of UI screenshots and identify which of two interfaces appears better. This screenshot-based setting is well suited for evaluating visual structure and perceived usability, but it does not capture whether an interface also supports reliable task completion in interactive use. For that aspect, dynamic assessment methods, such as computer-use agents that can interact with interfaces automatically, are particularly relevant.

Recent work on improving UI generation from designer feedback [**?** ] demonstrates that high-quality, workflow-aligned feedback can be far more informative than large amounts of coarse ranking data; notably, simple pairwise rankings in that study reached only around 50% evaluator agreement, close to chance level. These findings suggest that GUI quality assessment benefits from richer evaluation structures – such as rubric-based scoring and requirement-aligned judgment – rather than binary preference alone. In parallel, the broader LLM-as-a-judge literature [**?** ] has introduced prompting, consistency, and aggregation strategies that may also be useful for GUI evaluation. Taken together, these developments motivate a more methodologically explicit re-evaluation of GUI quality judgment with contemporary multimodal models.

This thesis therefore asks whether modern multimodal large language models can approximate human judgments of GUI quality beyond pairwise comparison, and how such judgments should be elicited and evaluated in a reliable way. It does so through a benchmark and leaderboard spanning a screenshot-based baseline track, a requirement-driven generation track, and a small dynamic module for task completion.

# 2 Planned Approach

## 2.1 Datasets and Evaluation Material

The evaluation material will be organized into two complementary tracks. The first track draws on publicly available screenshot-based resources from UIClip as an established baseline. These existing screenshots will be used for pairwise comparison and rubric-based assessment of visual design quality. To avoid evaluating only older interfaces, this material may be supplemented by a smaller curated set of contemporary web or mobile screenshots.

The second track will be constructed around natural language requirements. A set of short UI descriptions will be collected or designed, specifying the intended functionality, layout, and content of an interface. For this track, a small set of interfaces will be generated from shared natural language descriptions. Recent work has shown that current multimodal LLMs can generate renderable front-end code from visual and textual UI specifications [**?** ], making this pipeline practically feasible. The resulting front-end code will then be rendered as screenshots for visual evaluation while remaining executable for browser-based interaction tests and checks of whether the generated interface satisfies the original requirements.

## 2.2 Human Annotation Protocol

Human judgments will be collected for a selected evaluation subset. For pairwise tasks, annotators will compare GUI screenshots and indicate which interface is better overall, with an optional tie label if required. For single-image tasks, annotators will score each GUI with the rubric, providing brief justifications where appropriate. To improve the reliability of the reference labels, each item will be annotated by multiple raters, and inter-rater reliability will be measured, for example with Krippendorff's alpha.

For the dynamic evaluation module, assessment will rely primarily on automated task-completion metrics rather than full manual scoring. Limited human review may be used in cases where automatic outcome detection is ambiguous, both to verify whether a task was in fact completed and to distinguish interface-related failures from agent or evaluation-script errors. This review will also support a small qualitative categorization of major interaction failures, helping to explain where static quality judgments and dynamic task performance diverge.

## 2.3 Multi-dimensional GUI Rubric

To move beyond binary comparison, I will define a small rubric grounded in common visual design principles and usability heuristics. The initial dimensions are:

- **Layout quality**: spatial organization, alignment, spacing, and overall balance.
- **Visual hierarchy**: whether contrast, size, and color guide attention appropriately.
- **Information organization**: clarity of grouping and logical ordering of content.
- **Perceived usability**: how easily key actions and interface purpose can be inferred from a static screenshot.
- **Requirement fidelity** (Track B only): the degree to which the generated interface realizes the elements, structure, and functionality specified in the original natural language description.
- **Interaction quality** (Track B only): responsiveness to user actions, appropriateness of feedback and state transitions, and overall smoothness of the interactive experience when the generated interface is executed in a browser.

This list represents an initial set of dimensions; additional dimensions may emerge during pilot annotation and will be incorporated if they prove reliable and informative. Each dimension will be scored on a fixed ordinal scale with short written anchors so that the same rubric can be applied by both human annotators and model judges. The first four dimensions can be assessed from static screenshots alone, whereas requirement fidelity and interaction quality are specific to the requirement-driven track; the latter will be evaluated through the dynamic module in Section 2.6.

## 2.4 Prompting and Judging Strategies

The selection of judging strategies will be informed by an initial literature review of LLM-as-a-judge methods. In particular, this review will examine foundational work on pairwise judgment and judge bias [**?** ], broader taxonomies of LLM-as-a-judge techniques [**?** ], and recent evidence on position bias in pairwise evaluation [**?** ]. Based on this review, several judging strategies will be compared. A zero-shot baseline will ask the model directly for a pairwise preference or a rubric-based score. A few-shot variant will add one or two worked examples to test whether light calibration improves judgment quality. A rubric-guided condition will provide explicit scoring instructions and require a structured output format. To test robustness, the same prompt will be repeated to measure self-consistency. For pairwise tasks, order-swapped pairwise comparisons will be used, where the order of candidates is reversed to detect positional bias. In addition, the study will compare single-model judgments with simple aggregation strategies such as majority voting for pairwise tasks and averaged scores for rubric-based tasks.

## 2.5 Benchmark and Leaderboard Design

Beyond comparing individual models, the project will package the evaluation setup into a benchmark structure that combines pairwise tasks, rubric-based scoring, and shared reference annotations. Results will be stored in a standardized format so that newly added multimodal models can be evaluated under the same protocol by running the provided evaluation scripts against the shared benchmark dataset and submitting results in that format. As a practical artifact, the benchmark will be accompanied by a web-based leaderboard hosted on a platform such as Hugging Face Spaces, enabling transparent comparison of models. The leaderboard will report scores per rubric dimension and per task type—pairwise and rubric-based—allowing fine-grained comparison rather than a single aggregate score. This design ensures the benchmark remains easily extensible as new multimodal models become available.

## 2.6 Dynamic Evaluation with Computer-Use Agents

To complement the static benchmark, the project will include a targeted dynamic evaluation module. This module will operate primarily on the requirement-driven track, where LLM-generated HTML interfaces can be rendered and executed in a browser. For a selected subset of these interfaces, short interaction tasks derived from the original natural language descriptions will be defined and executed by computer-use agents. Task completion will be determined through automated outcome checks such as DOM state inspection or screenshot comparison against expected end states, with limited human review for ambiguous cases as described in Section 2.2. Prior work has demonstrated that multimodal agents can perform goal-directed interaction tasks on visual web interfaces [**?** ], providing a methodological foundation for this module. A key research question for this module is whether static quality scores are predictive of dynamic task success—specifically, whether interfaces that receive strong static quality judgments also support reliable task completion in practice, and whether requirement alignment in the static setting predicts successful interaction in the dynamic setting.

## 3 Evaluation

Several current multimodal models will be tested as judges, including representative models from major closed-source and open multimodal model families, depending on availability at evaluation time. As an exploratory comparison, the study may also include one or two smaller or weaker multimodal models in order to examine how judge quality varies with model capability. The pairwise setting will be evaluated through agreement with human majority labels, while the rubric-based setting will be evaluated through rank correlation and score differences between model outputs and human ratings. For the requirement-driven track, an additional evaluation dimension will measure how well model judgments capture whether a generated interface satisfies its original natural language description. The dynamic extension will be evaluated through task completion rate, action efficiency, and qualitative failure analysis on the selected interactive tasks. Reliability analysis will cover inter-rater agreement in the human annotations, consistency across repeated model judgments, stability under order-swapped pairwise comparisons, and agreement across different models. Computational cost will also be tracked in order to assess practical trade-offs between evaluation quality and API usage.

Finally, a qualitative error analysis will examine typical failure cases, such as cluttered layouts, minimal interfaces, text-heavy screens, and visually appealing but less usable designs. Together, these analyses should clarify how well modern multimodal large language models can serve as judges of GUI quality, which rubric dimensions are most reliable, and which judging strategies align best with human assessment.